DEENBANDHU CHHOTU RAM UNIVERSITY OF SCIENCE AND TECHNOLOGY, MURTHAL, SONIPAT

Summer Training Report on Web Development

Submitted in fulfillment of the requirements for completion of 6 th semester

of

Bachelor of Technology (CSE)

Submitted To:

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **INTERNSHALA** for providing me the opportunity to complete my **online internship in Generative AI**. It was a great learning experience that allowed me to apply my theoretical knowledge to realworld scenarios and gain valuable industry exposure.

I am deeply thankful to the **mentors and faculty at INTERNSHALA** for their constant guidance, encouragement, and support throughout the internship. I would also like to extend my appreciation to all the team members and coordinators for their cooperation, insightful feedback, and for making my learning process enriching and productive.

I am grateful to my university and faculty for their continuous support and for giving me the opportunity to undertake this internship as a part of my academic curriculum. This internship has been a significant milestone in enhancing my technical skills, understanding of generative AI concepts, and overall professional growth.

BHUPESH

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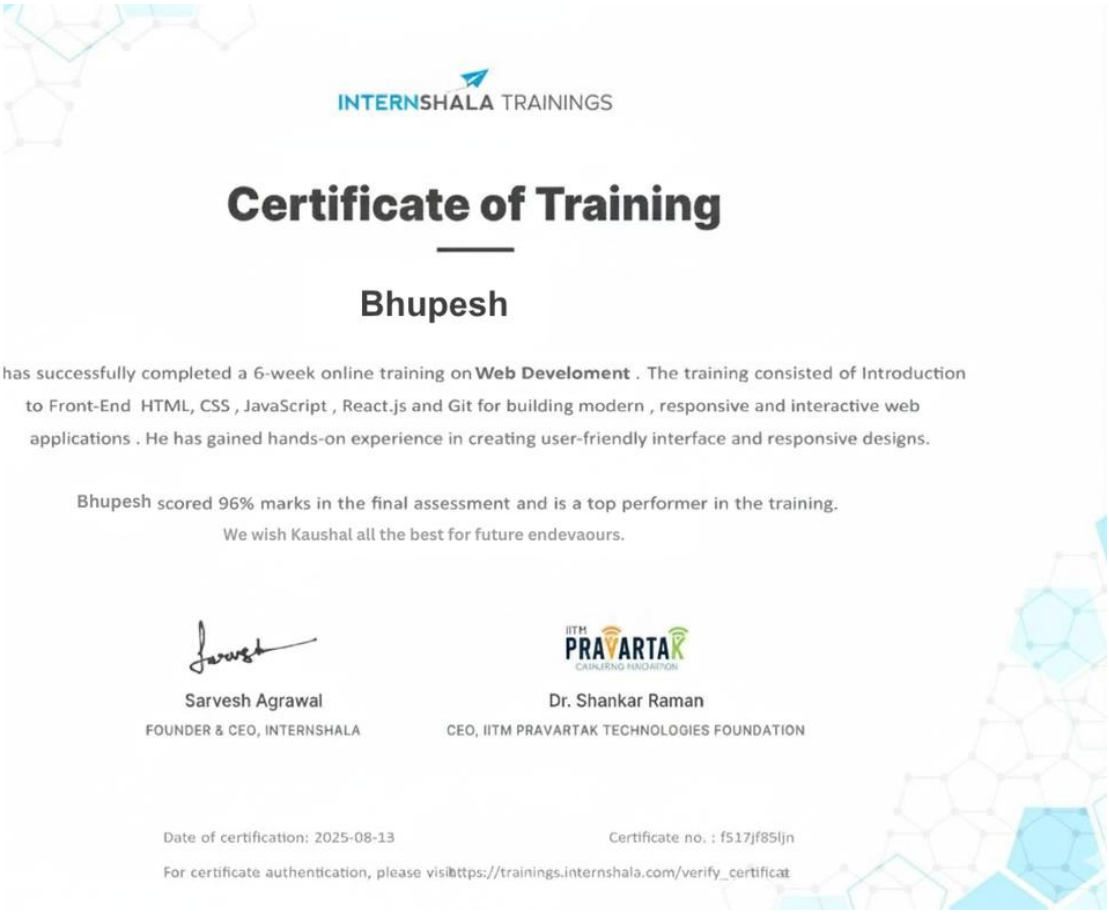
B.tech(CSE)

CERTIFICATE



INTERNSHALA TRAININGS

Certificate of Training

Bhupesh

has successfully completed a 6-week online training on **Web Develoment** . The training consisted of Introduction to Front-End HTML, CSS , JavaScript , React.js and Git for building modern , responsive and interactive web applications . He has gained hands-on experience in creating user-friendly interface and responsive designs.

Bhupesh scored 96% marks in the final assessment and is a top performer in the training.

We wish Kaushal all the best for future endevaours.

Sarvesh Agrawal
FOUNDER & CEO, INTERNSHALA

IITM PRAVARTAK CATALYSING FOUNDATION

Dr. Shankar Raman
CEO, IITM PRAVARTAK TECHNOLOGIES FOUNDATION

Date of certification: 2025-08-13

Certificate no. : f517jf85ljn

For certificate authentication, please visit https://trainings.internshala.com/verify_certificat

INTRODUCTION TO ORGANISATION

The Web Development Training by Internshala is a beginner-friendly online program designed to help students learn how modern websites and web applications are built from scratch. Its main objective is to teach learners how to build complete, real-world web applications using both frontend tools (including React) and backend technologies.

The course begins with the fundamentals of HTML, CSS, and JavaScript, which form the core of every website. Students learn how to structure webpages, style them, and add interactivity. Key JavaScript concepts such as variables, loops, functions, DOM manipulation, and event handling are explained in a simple, practical way, ensuring a strong foundation in programming.

The backend module covers Node.js and Express.js, where learners understand how servers work, how routing is handled, and how REST APIs are created. The training also includes MongoDB, teaching students how to store, update, and manage data for dynamic applications. By connecting the backend with the frontend, learners get a complete understanding of how full-stack web applications function in the real world.

Internshala also provides doubt-solving support, ensuring that learners get answers to their questions quickly. The training includes quizzes, assignments, a final test, and placement support such as interview preparation and access to internships. Overall, the Web Development Training offers a practical and comprehensive path for students to learn frontend and backend development, gain real-world experience, and build the skills needed for a career in web development.

TECHNOLOGIES OVERVIEW

The development of this web application is based on a modern full-stack architecture that incorporates powerful, efficient, and industry-standard tools. The chosen technology stack allows seamless integration between the frontend, backend, database, and deployment environment. The key technologies used in this project include React for building the frontend interface, Appwrite for backend services including authentication, database, and file storage, and Vercel for hosting and deploying the application. Each technology contributes significantly to the overall performance, scalability, and maintainability of the system.

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1. React (Frontend Technology)

React is an open-source JavaScript library used for building interactive, dynamic, and component-based user interfaces. Developed by Facebook (Meta), it is currently one of the most widely adopted frontend libraries across the tech industry. React enables the creation of reusable UI components, which improves development efficiency and reduces code repetition.

One of the core strengths of React is its virtual DOM mechanism. Instead of directly updating the browser's DOM — which can be slow — React updates a lightweight virtual copy of the DOM first. Only the parts that actually change are re-rendered, which significantly boosts performance and makes the user experience smoother.

React also offers powerful features such as hooks, which allow developers to manage states and lifecycle methods more easily. Hooks like useState, useEffect, and useContext simplify logic handling and enable cleaner code. The component-based architecture ensures better maintainability, modular code structure, and scalability of the application.

In this project, React is used to build the entire frontend interface, including navigation, dashboard views, file management screens, upload UI, and user interaction points. API calls to Appwrite services are

integrated directly from the React frontend, enabling real-time dynamic updates such as file uploads, folder creation, listing, renaming, and deletion. The use of React Router helps manage different pages and views within the application without reloading the entire site.

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2. Appwrite (Backend, Database, Authentication, and Storage)

Appwrite is an open-source backend-as-a-service (BaaS) platform that provides a complete backend solution without requiring developers to build everything manually. It supports essential backend functionalities such as user authentication, databases, cloud storage, serverless functions, permissions management, and APIs.

a. Backend Services

Appwrite manages server-side logic through its built-in REST APIs and SDKs. The application communicates with Appwrite's backend to perform actions like creating folders, uploading files, renaming items, deleting items, and retrieving user-specific content. This eliminates the need for building a backend server from scratch, reducing development time and complexity.

b. Authentication

Appwrite offers secure and customizable authentication mechanisms, including email-password login, OAuth support, session management, and JWT-based authentication. In this project, Appwrite handles all login and registration processes, ensures that user accounts are securely stored, and manages user sessions. This allows each user to have their private data space similar to cloud storage services.

c. Appwrite Database

The database provided by Appwrite is a highly scalable NoSQL document-based system. It allows structured storage of folders, file metadata, recent activity logs, user information, and any other necessary data. With built-in

permissions and role management, Appwrite ensures that users can access only their own resources.
The use of collections, attributes, and documents makes the database flexible and suitable for modern cloud applications. All data is fetched and updated through Appwrite's API, ensuring consistency between the frontend and backend.

d. Storage Services
Appwrite's storage module is used for handling all file uploads and media storage. Users can upload files, images, PDFs, documents, and other content, which is securely stored in the Appwrite cloud. Each file is associated with metadata and user-specific permissions. Appwrite's efficient handling of file uploads and retrieval enhances the overall reliability and usability of the application.

e. Real-Time Features
Appwrite also supports subscription-based real-time updates, meaning changes such as file uploads or deletions can be reflected instantly. This contributes to a smoother, dynamic user experience.

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3. Vercel (Hosting and Deployment)
Vercel is a cloud platform optimized for frontend frameworks, static sites, and serverless functions. It is widely used for deploying applications built with React, Next.js, Vue, and other modern frameworks. Vercel automates the entire deployment pipeline and provides a fast global content delivery network (CDN) to ensure high performance.

a. Deployment Process
Projects can be directly connected to GitHub or GitLab repositories. Whenever new changes are pushed to the repository, Vercel automatically builds and deploys the latest version of the application. This

continuous deployment approach ensures that updates are quick, consistent, and error-free.

b. Performance Optimization

Vercel delivers content using an edge network, which means users around the world experience minimal loading time. Features like automatic caching, asset optimization, compression, and image optimization further enhance performance.

c. Environment Variables & API Integration

Vercel allows secure configuration of environment variables, which is essential for connecting the React frontend with Appwrite backend securely. This ensures sensitive keys and credentials are not exposed publicly.

d. Scalability

Vercel automatically scales the application based on user traffic. Whether there are 100 users or 1 million, the platform adjusts resources to maintain speed and stability without manual configuration.

In this project, Vercel is used to host and deploy the React application. The deployment process ensures that the app remains accessible online with high performance and zero downtime.

PROJECT: StorX – Cloud based Storage Management System

1. INTRODUCTION

StorX is a cloud-based file storage and management system designed to provide users with a secure, organized, and intuitive platform for storing, accessing, and managing their digital files. Inspired by modern cloud platforms like Google Drive and Dropbox, StorX offers essential file-management capabilities combined with a clean user experience and efficient backend support. The project aims to simplify personal and professional file handling by giving users their own secure digital space, accessible from anywhere.

The system is built using a React-based frontend, powered by Appwrite for backend services such as authentication, real-time database handling, file storage, permissions, and user session management. The application is deployed on Vercel, ensuring fast global delivery and reliability.

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2. PROJECT OBJECTIVES

The main objectives of StorX include:

✓ **Providing a cloud-based file storage solution**

Users should be able to upload, store, view, and manage files in a secure online environment.

✓ **Ensuring user-specific isolation of data**

Each user gets their own separate storage space, ensuring privacy and data protection.

✓ **Offering a smooth and intuitive user interface**

The frontend is designed to provide a clean dashboard, easy navigation, and minimal complexity.

✓ **Supporting core cloud drive operations**

Features like folder creation, renaming, file uploads, deleting, restoring, recent history, and shared items are integrated.

✓ **Using modern and efficient technology**

React, Appwrite, and Vercel ensure high performance, scalability, and ease of maintenance.

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3. Key Features of StorX

StorX includes all essential features expected from a modern cloud storage platform:

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a. User Authentication

Users can register and log in securely using Appwrite's authentication system.

- Email-password registration
- Secure session handling
- User-specific access and permissions

---

b. MySpace (Personal File Repository)

This is the core section where users can view all their uploaded files and folders.

Features include:

- Uploading files of any type
- Viewing file list or grid
- Sorting and organizing files
- Creating folders to categorize items

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c. Folder Management

Users can create multiple folders to organize data.
Functions include:

- Folder creation
- Inline renaming
- Viewing folder contents
- Managing nested files

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d. File Upload System

Users can upload documents, images, PDFs, videos, audio, and more.
The upload system supports:

- Progress indication
- Metadata storage
- Automatic real-time updates
- File-size validation

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e. File Actions

Users can perform operations like:

- Rename files (extension preserved)
- Delete files
- Restore files from Bin
- Download option (if implemented)
- Preview (if implemented or supported for specific files)

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f. Recent Activity

A dedicated section lists recently uploaded or modified files, making it easier for users to find their latest work.

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g. Shared With Me

This section stores files shared by other users (if sharing functionality is implemented).
Even if not implemented fully, the design accommodates future expansion.

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h. Bin (Recycle Bin)
Deleted files are not removed immediately; instead, they are moved to the Bin.
Users can:

- Restore deleted files
- Permanently delete files

This provides an extra layer of safety against accidental deletion.

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4. System Architecture

StorX follows a **front-end + backend-as-a-service** architecture.

A. Frontend (React)
The React client communicates with Appwrite using its JavaScript SDK.
Main responsibilities of frontend:

- Rendering UI components
- Fetching data from Appwrite
- Showing real-time updates
- Handling file uploads and user interactions
- Routing between dashboard sections
- Managing app state using hooks and context

B. Backend (Appwrite)
Appwrite handles all server-side responsibilities, including:
1. Authentication

- Manages user login/registration

- Generates secure sessions
- Restricts database/file access based on user ID

2. Database

Used to store metadata:

- File details
- Folder structure
- Recent activity logs
- User references
- Bin records

Appwrite's document-style structure ensures fast read/write operations.

3. Storage

Used for storing actual file content.

Each uploaded file is saved in buckets with access permissions.

4. Real-time Services

Changes such as file uploads and folder creation can trigger immediate UI updates.

C. Deployment (Vercel)

The React application is hosted on Vercel, which:

- Provides serverless hosting
- Ensures fast performance due to global CDN
- Automates CI/CD deployment from GitHub
- Manages environment variables securely
- Offers automatic scaling during traffic spikes

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5. User Workflow

Step 1: Registration/Login

User creates an account or logs in via Appwrite authentication.

Step 2: Dashboard Access

They are redirected to the StorX dashboard showing:

- MySpace
- Recent
- Bin
- Shared With Me

Step 3: File/Folder Actions

User performs tasks like:

- File upload
- Folder creation
- Navigation inside folders
- Renaming items
- Deleting or restoring files

Step 4: Data Synchronization

The app communicates with Appwrite to store files and metadata, which update the UI in real-time.

Step 5: Logout

User can log out anytime, and a new session must be created for access.

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6. Design & User Interface

The UI is designed to be simple, minimalistic, and user-centered.

Key UI components include:

- Sidebar navigation menu
- Top bar with user profile
- File cards/grid view
- Folder view
- Upload modal
- Rename input field
- Toast notifications

Color themes, icons, spacing, and layout contribute to a professional experience similar to cloud storage apps.

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7. Security Considerations

StorX incorporates:

- User-based permission rules
- Protected API endpoints
- Secure file access (files only accessible by the owner)
- Safe session handling from Appwrite
- No direct exposure of tokens or keys

Data isolation ensures zero risk of users accessing each other's files.

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8. Advantages of StorX

- Fast and smooth performance due to React and Vercel
- Strong and secure backend with Appwrite
- Organized file management
- User-friendly interface
- Scalable architecture
- Easily extendable for future upgrades

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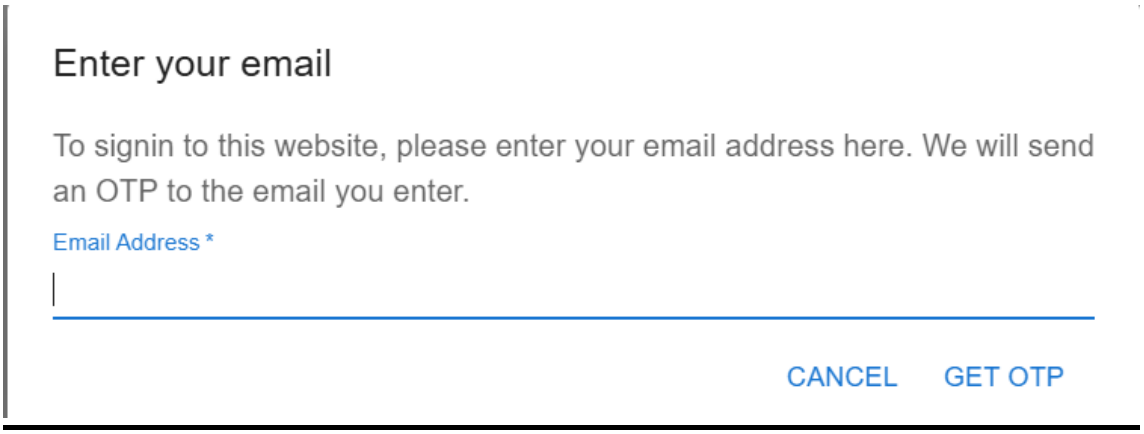
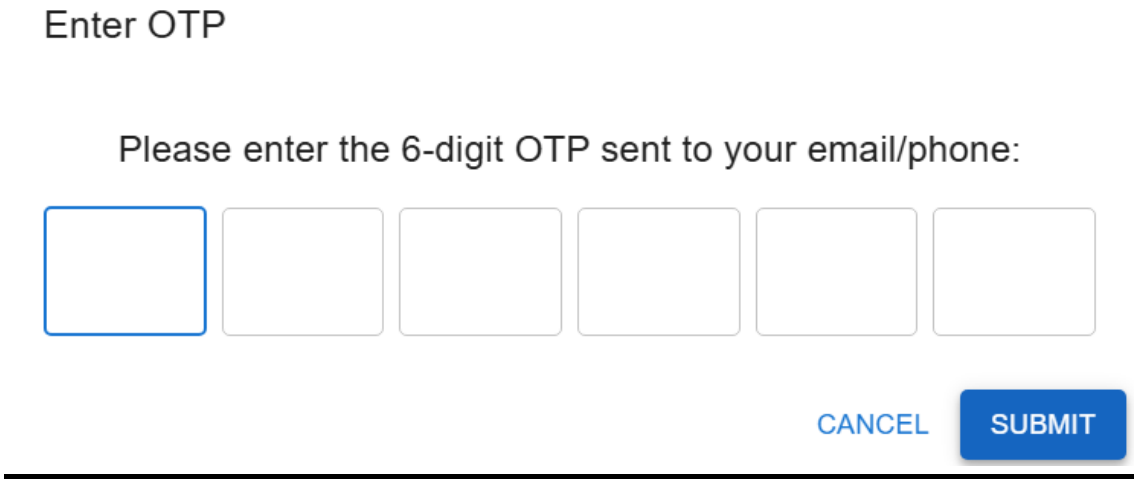
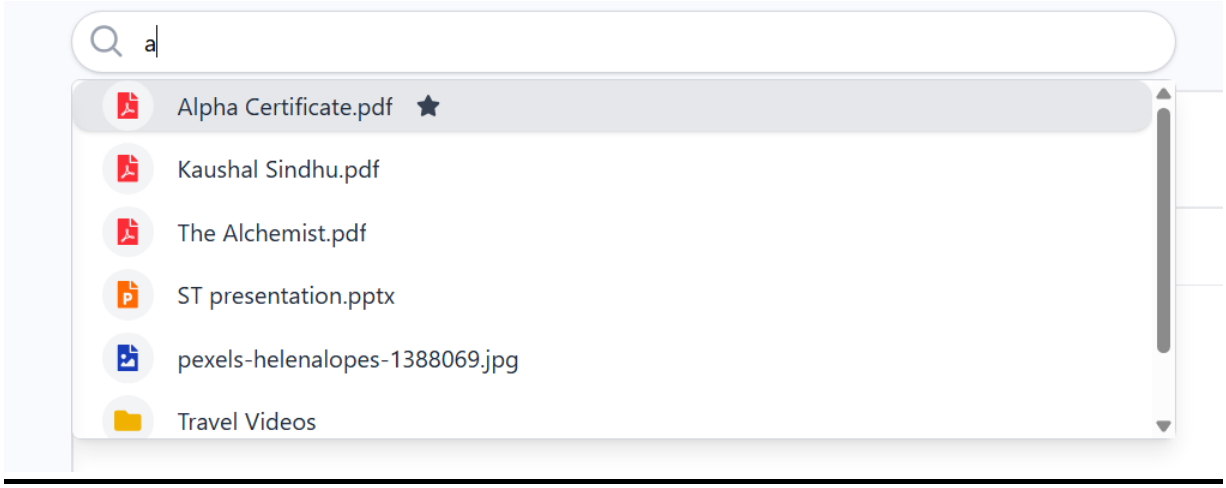
9. Possible Future Enhancements

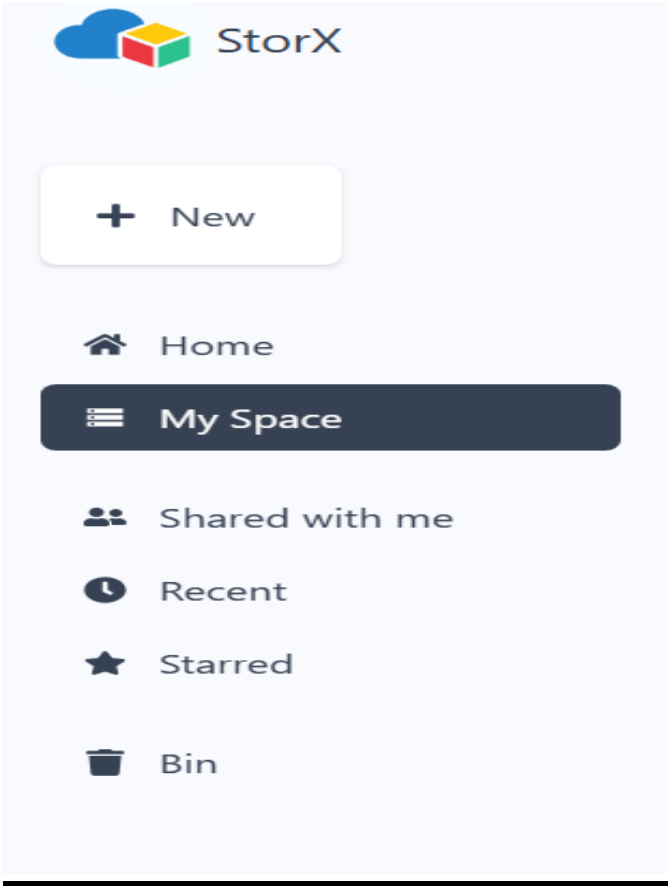
- File sharing with permissions
- Real-time collaboration
- File previews (PDF, images, docs)
- Mobile app version
- Notifications and email alerts
- Storage limit visualization
- Admin dashboard

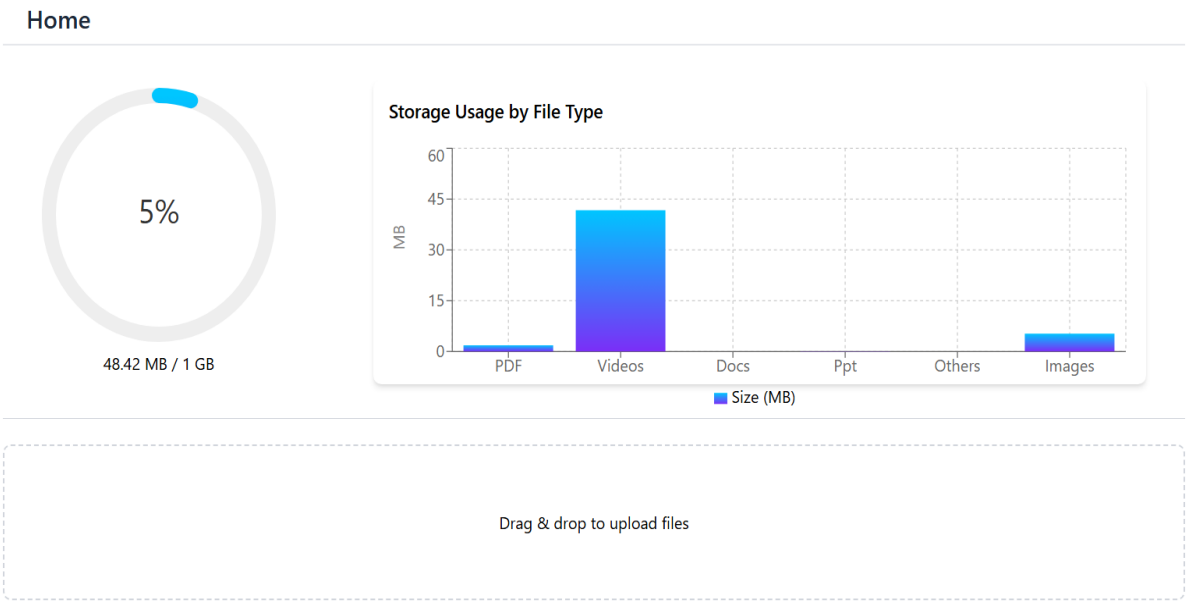
PROJECT VISUALS








StorX
New
Home
My Space
Shared with me
Recent
Starred
Bin


Home
5%
48.42 MB / 1 GB
Storage Usage by File Type
MB
60
45
30
15
0
PDF
Videos
Docs
Ppt
Others
Images
Size (MB)
Drag & drop to upload files

CONCLUSION

The development of StorX – Cloud Storage Management System marks a comprehensive exploration of modern web development practices, combining frontend, backend, database, and cloud deployment technologies into a single, fully functional application. Throughout the project, the use of React for the user interface, Appwrite for backend services, and Vercel for deployment enabled the creation of a scalable, secure, and responsive cloud-storage platform that closely resembles real-world commercial systems.

StorX successfully provides essential file-management features such as file uploads, folder creation, inline renaming, deletion and restoration, recent activity tracking, and user-based data isolation. These capabilities demonstrate a strong understanding of full-stack development concepts, API integration, cloud storage handling, and user experience design. The project also highlights efficient handling of authentication, database operations, and cloud storage mechanisms through Appwrite's powerful backend services.

This project not only meets its initial objectives but also opens doors for future enhancements such as file sharing, real-time collaboration, previews, and advanced storage analytics. The experience gained through building StorX has strengthened concepts in React development, backend integration, system design, and scalable deployment practices.
Overall, StorX stands as a complete, practical, and professionally executed web application that reflects the ability to build modern cloud-based systems and offers a strong addition to the developer's portfolio.